

Agentic AI Engineering

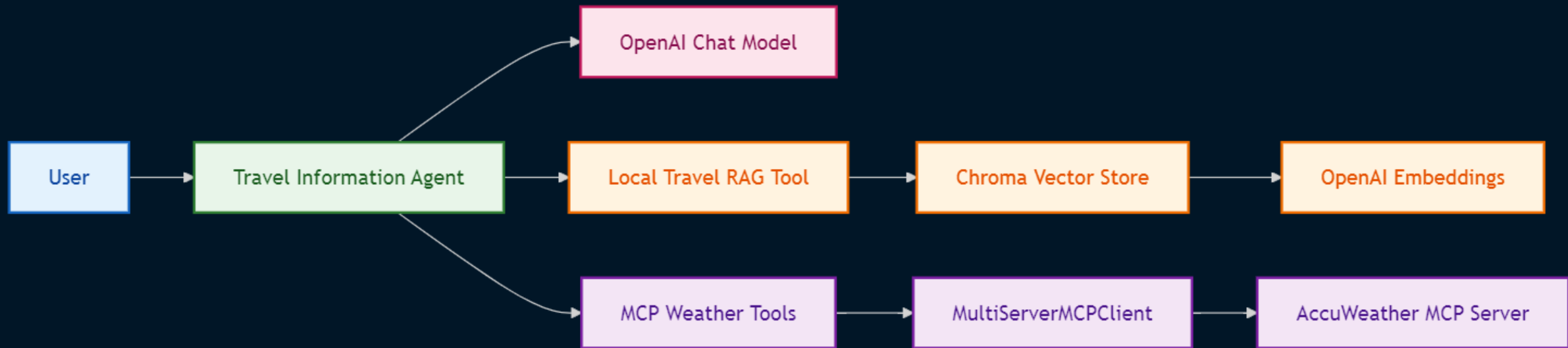
MCP-Enabled Agentic AI Assistant

Local RAG + Remote Tool Orchestration

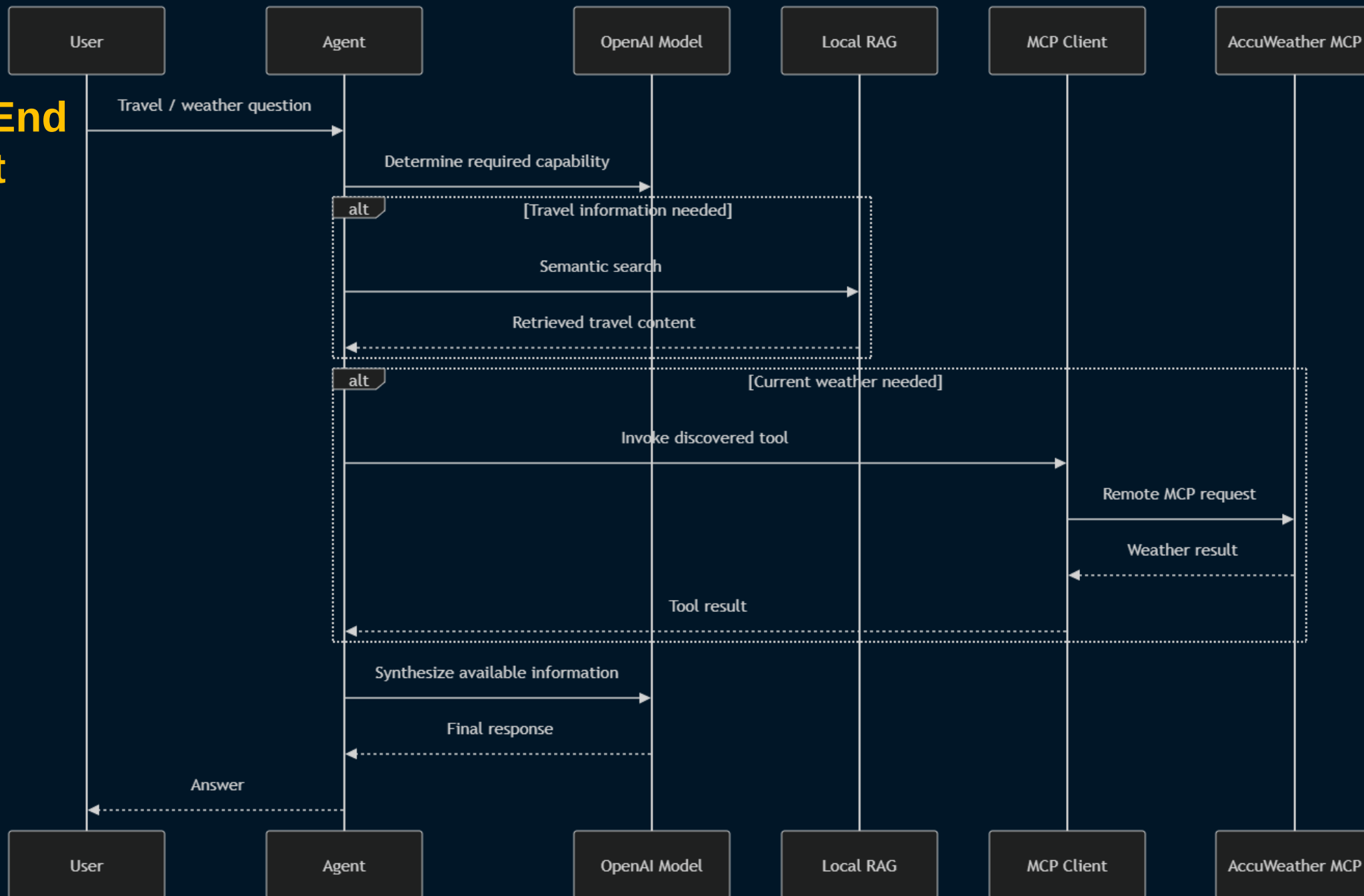
Max Montaseri

- Architected an Agentic AI assistant combining **local RAG with dynamically discovered remote MCP tools**, demonstrating protocol-based integration between LLM reasoning and external services.
- Integrated MultiServerMCPClient with a streamable HTTP AccuWeather MCP server to support **dynamic remote tool discovery and invocation**.
- Engineered asynchronous agent execution using **Python AsyncIO, LangChain agents, MCP adapters, and ainvoke()** for network-bound tool workflows.
- Built a local semantic-retrieval pipeline using **OpenAI embeddings, Chroma, asynchronous HTML ingestion and recursive document chunking**.
- Unified local and remote capabilities into a single agent toolset, allowing the LLM to select between knowledge retrieval and live weather information.
- Implemented typed agent state using TypedDict and separated configuration, retrieval, MCP integration, tool composition, and agent execution into modular components.
- Validated remote tool discovery and invocation through interactive testing, terminal inspection, and LangSmith trace verification.

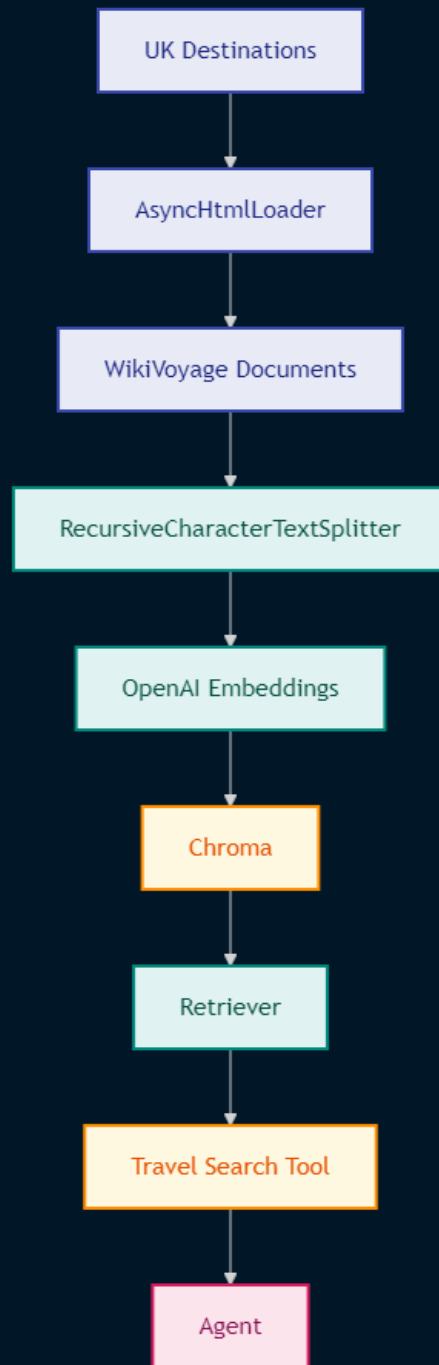
Architecture



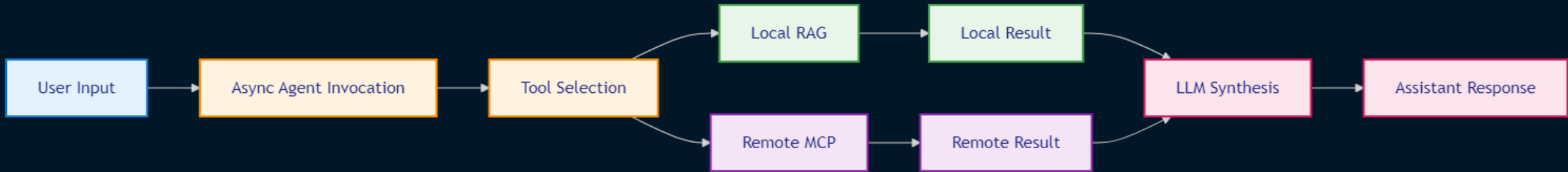
End-to-End Request Flow



Local RAG Architecture



Asynchronous Execution Runtime Model



Project Flow

